TEAM CUPIQ · EGYPTIAN CHINESE UNIVERSITY

VitalOrbit

An autonomous onboard health system for deep-space crews.

When Earth is **19 light-minutes** away, the crew is its own first responder.



THE PROBLEM

On Mars, help can't be 40 minutes away.

Today, astronaut health depends on flight surgeons on the ground. On a Mars mission that link breaks down.

4–24 min

one-way signal delay

A question and its answer take up to 48 minutes; in our mission model Earth is 19.1 light-minutes away on day 432.

~2 weeks

solar conjunction blackout

When the Sun sits between Mars and Earth, there is no link at all.

5 hazards

NASA's RIDGE risks

Radiation, Isolation, Distance, Gravity (bone, muscle, vision), hostile Environment: slow damage that no one on Earth sees in time.

THE SOLUTION

One system that detects, guides and remembers.

Personal anomaly detection

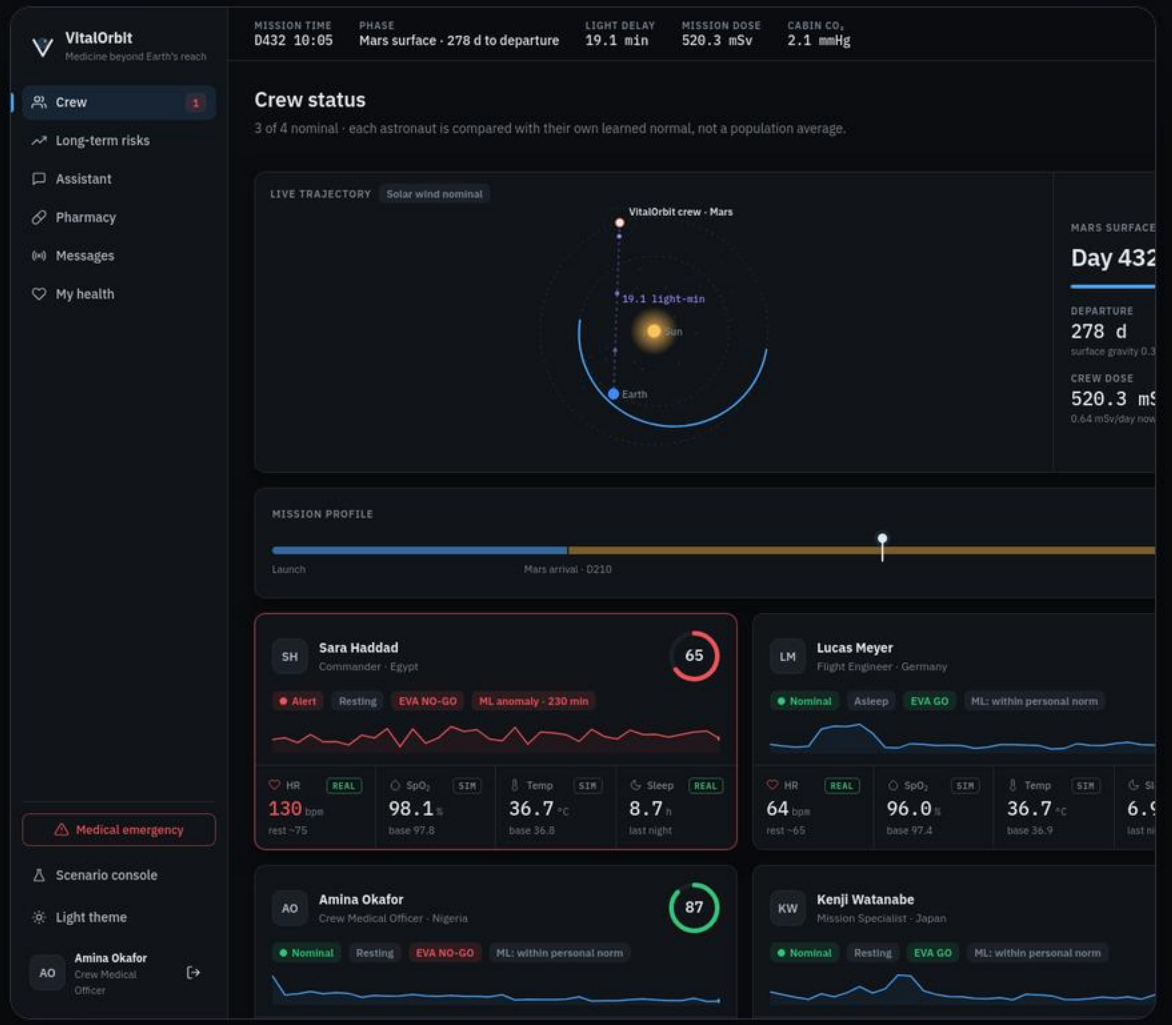
An ML model per astronaut, trained on their own real wearable data, plus clinical safety limits.

Long-mission risk forecasting

Radiation vs career limit, bone and muscle loss, vision (SANS), mood: projected to mission end.

Offline medical assistant + smart pharmacy

Guided first aid with timers, voice and Arabic; every drug checked for stock and radiation-shortened expiry.



HOW IT WORKS

From a heartbeat to a decision, onboard.

REAL SIM

Vitals

Fitbit heart rate, activity, sleep (public dataset); simulated SpO_2 and temperature.

→

Personal ML

Isolation Forest on deviations from each person's own rhythm; alert after 30 min of persistence.

→

Check-in

"How do you feel?" The answer labels the alert and opens the right protocol.

→

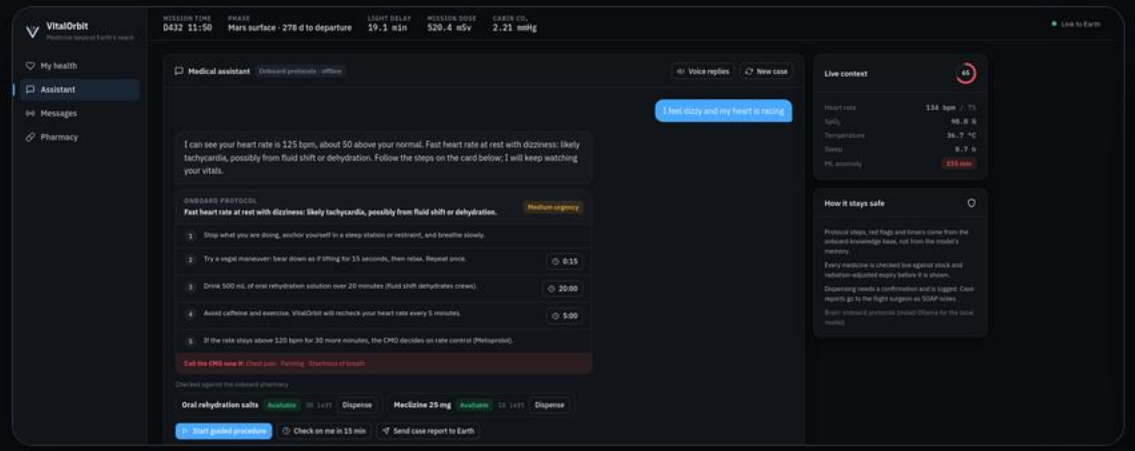
Assistant + pharmacy

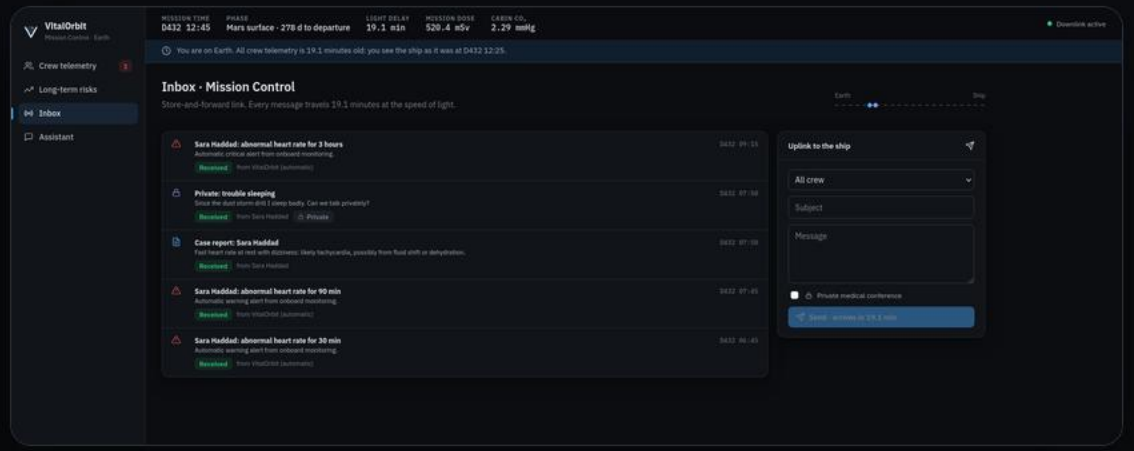
Steps, timers, red flags; medicine checked live. Dispensing is logged.

→

Earth, delayed

SOAP case report and alerts arrive 19 min later; private line to the flight surgeon.





DOES IT WORK?

Measured on real wearable data.

3,456 real readings from 4 people, plus 12 injected emergencies (tachycardia, low oxygen, fever).

12 / 12
emergencies detected, correct vital named every time

45 min
median time to the first alert, critical within 3 h in 100 %

0.08
false warnings per crew-day (1 in 12 crew-days)

0
false critical alerts

Honest by design

Every number in the app is tagged REAL, SIM or MODEL.

- Fitbit dataset, Zenodo 53894 (CC-BY 4.0)
- Radiation: Curiosity RAD, 1.8 mSv/day cruise, 0.64 on Mars
- Career limit 600 mSv (NASA-STD-3001)
- Bone loss and countermeasures: LeBlanc 2000, Smith 2012
- Drug potency in flight: Du et al. 2011
- Light delay and conjunction from an orbital model

IMPACT

Built for Mars. Useful wherever the doctor is far away.

For crews

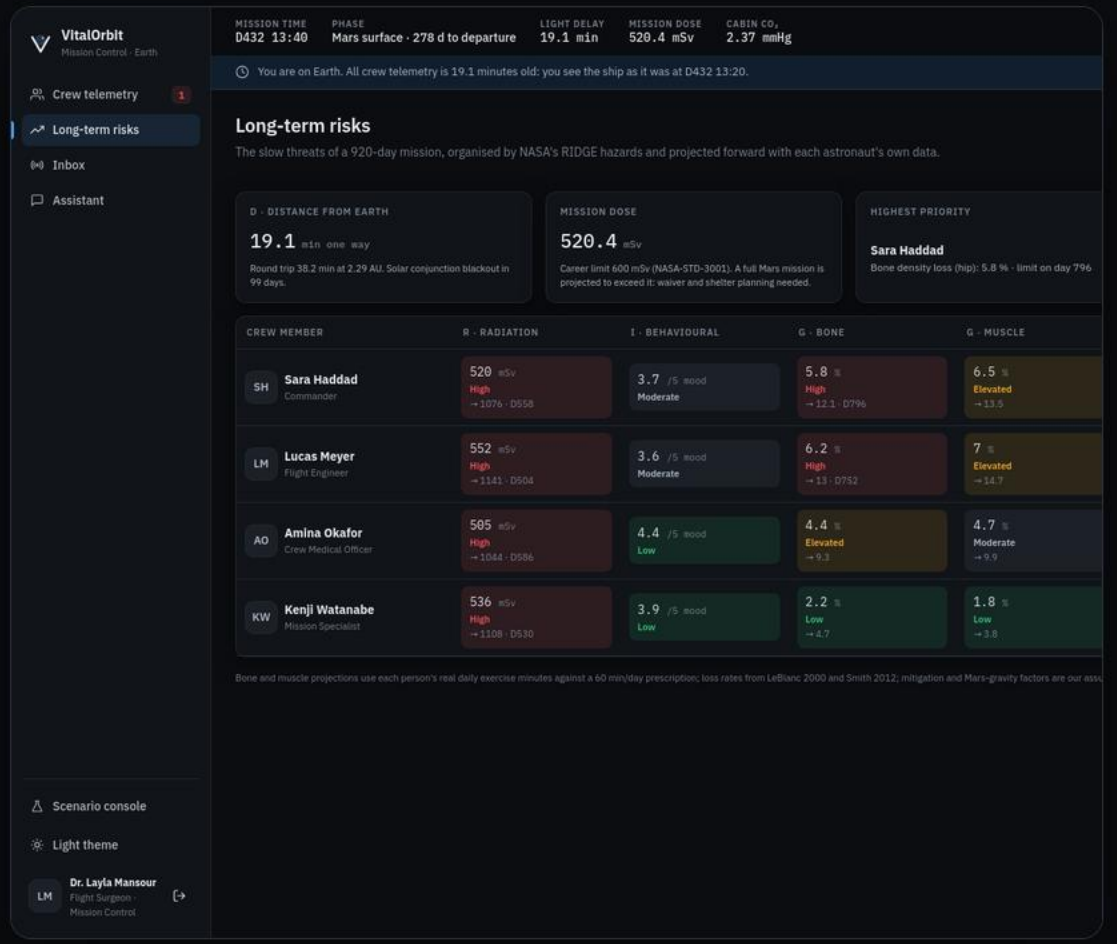
Earlier warnings, guided care in minutes instead of a 40-minute round trip, privacy for each astronaut.

For flight surgeons

Structured SOAP reports and long-term risk trends, ready when the link returns.

On Earth

Antarctic stations, submarines, offshore rigs, remote villages and field hospitals.



THANK YOU

Medicine beyond Earth's reach.

Code and run instructions
github.com/3AMURR/VitalOrbit

Run it: `./run.sh` · self-running demo: `localhost:8000/?tour=1`



Team CUPIQ · Egyptian Chinese University · Built with help from Claude (Anthropic)